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Population genomic analyses reveal extensive genomic regions within selective sweeps associated with adaptation and demographic history of a wheat fungal pathogen

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Abstract

Plant pathogens rapidly evolve in response to changes in their hosts and environments, resulting in destructive epidemics. The fungus *Puccinia striiformis* f. sp. *tritici* (*Pst*), responsible for wheat stripe rust disease, exemplifies such evolutionary dynamics. Host population changes impose selective pressures on pathogens; however, the selection footprints and demographic patterns of *Pst* are not well understood. In this study, we performed whole-genome sequencing of 69 *Pst* isolates collected from diverse regions across China to investigate genome-wide patterns of gene flow and divergence. Our findings reveal significant gene flow within China but limited exchange between Chinese and international *Pst* populations, characterized by a slower decay of linkage disequilibrium compared to fungi with known sexual reproduction. We identified extensive hard and soft sweeps associated with *Pst* adaptation. The genes involved in these sweeps are enriched for secreted proteins and effectors, with functions related to pathogenicity, temperature tolerance, and fungicide resistance, indicating positive selection driven by both host and abiotic pressures. Demographic reconstruction shows strong bottlenecks in *Pst* populations during the domestication of wheat approximately 10,000 years ago and the advent of modern agriculture 100 years ago, suggesting that crop domestication and breeding programs have significantly influenced pathogen population dynamics. Our study provides valuable insights into the genomic

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evolution of *Pst* in China and highlights the impact of modern agricultural practices on pathogen demography.

KEYWORDS
demographic history, population genomics, *Puccinia striiformis* f. sp. *tritici*, selective sweeps, wheat stripe rust

1 | INTRODUCTION

Fungi are ubiquitous and play fundamental roles in ecosystems. Among them is a group of pathogenic fungi that cause severe diseases in plants and pose a threat to worldwide food security.[1,2] These invasive plant fungal pathogens often exhibit the ability to rapidly evolve in response to novel environments, particularly in agricultural ecosystems.[3–6] Rapid evolution can lead to the emergence of diseases in new host species or cultivars, or the reemergence of pathogens with increased virulence. In either case, these pathogens become more adapted, increasing the potential for widespread epidemics. Understanding the biological and evolutionary features underlying invasive plant fungal pathogens is vital for the implementation of effective control strategies and is a key focus within the fields of population genetics and molecular ecology.[7]

Puccinia striiformis f. sp. *tritici* (*Pst*) is a fungal pathogen of wheat that causes stripe rust disease, one of the most destructive wheat diseases worldwide, especially in China, the United States, and Europe.[4,8] *Pst* is a heteroecious fungus, necessitating two taxonomically distinct host species to complete its life cycle. The asexual stage, which is the predominant phase, occurs on wheat (*Triticum aestivum*) and involves the production of urediniospores. Severe epidemics of wheat stripe rust are attributed to the asexual urediniospores, as multiple cycles of asexual reproduction can be completed within a single wheat growing season. The sexual stage, which has only recently been identified, takes place on *Berberis* and *Mahonia* species.[9,10] Although tens of *Berberis* and *Mahonia* species could serve as alternate hosts for *Pst,* the frequency of sexual reproduction in natural populations appears to be low, partially due to the incompatible phenology of these hosts.[11] The frequency of sexual reproduction in *Pst* natural populations remains unclear. The most effective strategy for controlling this disease is the deployment of resistant wheat cultivars.[12] However, similar to other invasive fungi, *Pst* can evolve rapidly, resulting in the emergence of new aggressive races (also known as pathotypes). These new races often render widely grown resistant cultivars ineffective within just a few years after their release.[5,13–15] In China, a significant rise in the occurrence of race CYR33, which exhibits virulence toward *YrSu*, was the predominant virulence shift observed from 2003 to 2007. Although certain cultivars retained resistance to CYR32 and CYR33, such as Guinong 22, which harbors *Yr24*/*26* gene in major regions of Sichuan Province, a novel virulent pathotype, CYR34, emerged in 2009. By 2015, CYR34 had emerged as the primary virulence source against cultivars containing the the *Yr24*/*26* gene.[16,17] Globally, several documented epidemics have occurred due to the emergence of new *Pst* races, including the breakdown of *Yr17* resistance in the 1990s and the emergence of the "Warrior" race in northern Europe in 2011.[18] So far, a comprehensive understanding of the evolutionary processes underlying the rapid evolution of *Pst* remains elusive.

Survival in harsh environments (e.g., scorching summers and chilly winters) is critical for plant pathogens and subsequent disease development. For *Pst*, oversummering and overwintering are two bottlenecks in its disease cycle. Although the wheat-growing areas in China exhibit great variation in geomorphological and climatological features, the pathogen is capable of both oversummering and overwintering across different regions. Therefore, it is proposed that wheat stripe rust has an independent epidemic system in China.[8,19] Based on geographic distribution, cropping systems, extensive field surveys, and long-term monitoring, the main areas where stripe rust is prevalent in China are divided into three epidemiological regions. The first region mainly covers the provinces of Yunnan, Chongqing, and Guizhou and their neighboring areas, where *Pst* can continually multiply during the winter and serve as sources of inoculum for local and northern regions in the following spring. This region is designated as the spring sources of inoculum (SprS) region. The second epidemiological region, referred to as the autumn sources of inoculum (AutS) region, comprises southern Gansu, northwest Sichuan, and the adjacent regions. The third epidemiological region, termed the spring epidemic areas (SprE), includes the wheat-growing areas of the Huang-Huai-Hai Plain and the lower reaches of the Yangtze River.[8] On a global scale, it is believed that the likelihood of population exchange between *Pst* populations in China and those in other countries is minimal due to the geographical barriers.[20] However, wheat stripe rust is an airborne disease, and its urediniospores can be dispersed by wind over long distances. Even though few studies have been conducted to examine the population diversity and differentiation between *Pst* populations from China and other countries, and only a limited number of isolates and

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molecular markers from China have used used,[21,22] the potential for admixture of *Pst* populations on a global scale needs to be further evaluated.

During a successful infection, pathogens have to overcome several layers of the host immune systems.[23] In the first layer, microbial/pathogen-associated molecular patterns (MAMPs/PAMPs) are detected by plant pattern recognition receptors, triggering pattern-triggered immunity (PTI) in plants. The pathogens then deliver effectors (typically small secreted molecules) to suppress PTI, resulting in the susceptibility of plants.[24] In the second layer, these secreted effectors (or their action) are recognized by the cell-internal immune receptors, often the products of resistance genes in the plant, leading to effector-triggered immunity (ETI). Consequently, due to the evolutionary pressures caused by such resistance genes (PTI), pathogens that acquire new effectors, which are no longer recognized by the immune receptors (due to mutations), or have lost the immune-triggering effectors, will be selected in the population, allowing them to overcome plant ETI. In this arms race between pathogens and hosts throughout evolution, changes in host populations exerted continuous selective forces on the dynamics of pathogens.[23,25] Such coevolutionary dynamics provide valuable opportunities to detect the footprints of pathogen adaptation within genomes.[26–34] Specifically, in the genome of *Pst*, hundreds of effector-coding genes have been annotated; however, the functions of most remain unknown. Therefore, the identification of candidate effectors under selection will be valuable for both functional studies and for understanding the rapid adaptation of *Pst*.

The impact of crop domestication on pathogen evolution remains poorly understood. Crop domestication has often resulted in a significant reduction of the genetic variation in the crop population.[35,36] The reduction may cause bottleneck effects for pathogen populations during crop domestication. Recently, high-throughput genomic sequencing has enabled a more detailed inference of population demography.[34,37–40] *Pst* is an obligate biotroph, meaning that it can only infect and obtain nutrients from living tissue. Such an intimate association provides a unique opportunity to study the demographics of *Pst* during wheat domestication. This will enhance our understanding of pathogen population changes in response to host resistance strategies employed in modern agriculture.

Here, we aimed to reveal the evolutionary features underlying the rapid adaptation of the *Pst* pathogen on wheat. We performed whole-genome sequencing (WGS) on 69 *Pst* isolates collected from various regions across China in 2015. Besides, we retrieved 28 published *Pst* genome sequences from known geographical distributions, including India, Europe, the US, Australia, and Eastern Africa, to represent the international population of this species. We then applied population genomic analyses to address the following key questions: (1) What is the genetic makeup of *Pst* populations? Does migration or admixture exist between *Pst* populations from China and other countries? (2) What is the extent of sexual reproduction in *Pst*, especially compared with its sister species, *P. graminis* f. sp. *tritici*, which is known for its sexual reproduction? (3) Are there footprints of selective sweeps in the genome of *Pst*? Do these genomic regions involve effector genes that are associated with adaptation? (4) Can we infer the demographic history of *Pst*? How is this affected by wheat domestication? This information will enhance our understanding of the adaptation of this destructive pathogen and allow us to evaluate the potential impact of modern agriculture on the evolution of pathogens.

2 | MATERIALS AND METHODS

2.1 | Isolate collection

We selected 69 *Pst* isolates collected in 2014 or 2015 from the provinces of Anhui, Chongqing, Gansu, Guizhou, Hubei, Qinghai, Shaanxi, Sichuan, and Yunnan, covering the major wheat production areas across China (Table S1). After purification from single uredinia, fresh urediniospores were used to inoculate Mingxian 169, a wheat variety known to be susceptible to all Chinese *Pst* races, for multiplication. The inoculation procedure was previously described by Liu et al.[41]

To explore the population structure of Chinese *Pst* populations in relation to international *Pst* populations, sequencing data of 28 worldwide isolates published on NCBI were also retrieved. These isolates were from Australia, Denmark, Ethiopia, Eritrea, Europe, France, India, the UK, and the US.[42–48] All isolates were collected from wheat. It should be noted that the international isolates were sampled between 1978 and 2011, whereas our Chinese isolates were collected between fall 2014 and spring 2015. Given that *Pst* was mostly clonal and there was no observed population stratification based on sampling dates (see RESULTS), the difference in sampling dates between the international and Chinese isolates is unlikely to affect our analyses.

2.2 | DNA extraction and WGS

Genomic DNA was extracted from urediniospores using the CTAB method.[49] The quality was assessed using a 0.8% agarose gel and an ND-1000 spectrophotometer (Bio-Rad, Hercules, CA, USA). Only high-quality genomic DNA, characterized by an A260/280 ratio of 1.8–2.0 and the absence of contaminating substances,

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was used as input material for library preparation. Whole-genome sequencing was conducted using the Illumina NovaSeq 6000 platform with a 150-bp paired-end module at Sinobiocore Biotechnology Co. Ltd.

2.3 | Reads processing and genomic variant detection

The sequencing supplier provided both raw reads and clean reads for each isolate. The clean reads were generated from the raw reads by discarding contaminations or low-quality reads based on the following criteria: (a) adapter contamination; (b) more than 10% "N" bases in either read; and (c) the proportion of low-quality bases (Phred quality < 5) exceeding 50% in either read. The Illumina clean reads of each isolate were further trimmed by removing adapter and primer sequences from the ends using Trimmomatic-0.39, with the following settings: LEADING:20, TRAILING:20, SLIDINGWINDOW:4:20, and MINLEN:100.[50] The high-quality trimmed reads were then mapped to the *Pst* CYR34 reference genome using the BWA-MEM with default parameters, except for –r 1.0.[51,52] The CYR34 genome was selected as the reference because it is the most prevalent race in China, and it is well annotated with high quality, with the BUSCO completeness of 97.0%. After removing duplicates (the identical reads mapped to the same genomic region, usually generated from library amplification and could result in false positives in variant calling) using SAMtools v0.1.19, the SAM-formatted alignments were converted to BAM format.[53] The Picard Tools software (http://picard.sourceforge.net/) was used to clean and sort the BAM files, which were then indexed using SAMtools. To obtain a preliminary quality control before variant calling, Qualimap v2.2.1 was used to check the mapping coverage and quality of all isolates.[54]

To reduce false positives and mismatched variants in the alignments, we conducted the GATK HaplotypeCaller twice following the recommended procedures of GATK 4.1.[55] We performed three steps to obtain high-quality known sites. First, we conducted an initial round of variant calling using GATK HaplotypeCaller to identify false-positive variants. Then, we used VCFtools v0.1.16 to filter biallelic SNPs and InDels with a minimum quality threshold of 1000.[56] After indexing the VCF file, a base quality score recalibration step was conducted to recalibrate the data. Then, the high-quality known sites were subjected to the second round of variant calling using GATK HaplotypeCaller for each isolate.

GVCF files of different isolates with only bialleles and a minimum variant calling quality of 1000 were acquired using VCFtools. To combine the multisample GVCFs, we performed GATK CombineGVCFs and GenotypeGVCFs and obtained a final joint genotyping VCF file containing all variants of all isolates. Then, we used GATK SelectVariants to separate SNPs and InDels. We only kept high-quality SNPs genotyping within coverage from 20× to 120× to reduce false positives in repetitive regions. Given that rare alleles may cause false-positive results in the following analyses, we removed variants with a minor allele frequency less than 0.05. These filters were implemented using VCFtools with the parameters set as –min-meanDP 20, –max-meanDP 120, –minQ 1000, –minGQ 40, –maf 0.05, and –max-missing 0.8.

2.4 | Population structure and summary statistics

We analyzed the population structure of 97 worldwide *Pst* isolates using two methods: principal component analysis (PCA) and ADMIXTURE. To avoid biases caused by linkage disequilibrium (LD), the SNPs used for detecting population structure were filtered using the "–thin 1000" option in VCFtools. After creating the BED file with PLINK, we used the GCTA software to perform PCA on all *Pst* isolates.[57] We set the PC number to 10 and selected the top three principal components to draw a scatter plot with the R package ggplot2. ADMIXTURE v1.3.0 software was used to analyze the relationships among all isolates, with K values set from 2 to 10.[58] For each group, we performed 50 iterations to compare the cross-validation (CV) errors of each K value. The K value with the lowest CV error was set as the most suitable population number. An isolate was assigned to the specific cluster when it had at least 70% ancestry from that cluster. Otherwise, it was considered an admixture. The admixed isolates were excluded from subsequent demographic estimates and analyses of selective sweeps. To explore the genetic composition of populations from each epidemiological region and province, we plotted a Sankey plot, a type of data visualization showing the flow of information (e.g., number of isolates) between different variables (e.g., epidemiological regions, provinces, and genetic clusters).

Genetic differentiation index (*Fst)* and nucleotide diversity (θ_π) for each of the seven populations across 10 kb windows of the genome were calculated using VCFtools. To detect the departure from the standard neutral model, we calculated Tajima's *D* statistics at both the genome-wide level and genomic windows along each chromosome, also using VCFtools.[59] A Tajima's *D* value of zero suggests that the population evolves under an equilibrium of mutation and genetic drift, with no selection at a specific locus. Positive values of Tajima's *D* (with low levels of both low- and high-frequency polymorphisms) indicate a decrease in population size or balancing selection, whereas negative values (with an excess of low-frequency

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polymorphisms) suggest population size expansion or directional selection.

We calculated LD using VCFtools, with the –geno-r2 and –ld-window-bp 20,000 options, for the Chinese *Pst* population and the international *Pst* population excluding China. The rate of LD decay could indicate the level of sexual reproduction. For comparison, we also reanalyzed our previously established *Pst* selfing population with 93 isolates.[60] In addition, we retrieved a recently available whole-genome dataset of *Puccinia graminis* f. sp. *tritici* (*Pgt*), the sister species of *Pst* and the causal pathogen of wheat stem rust disease.[61] It is well known that the *Pgt* fungus undergoes frequent sexual reproduction under natural conditions.[62,63] Therefore, it can serve as a reference for evaluating the extent of sexual reproduction in rust fungi.

2.5 | Genome regions scanning for selection

A selective sweep describes the phenomenon in which a beneficial variant arises and increases to high frequency. Generally, there are two kinds of selective sweeps depending on the number of advantageous haplotypes in a population: hard sweeps, characterized by a single selectively advantageous haplotype with high frequency, and soft sweeps, which involve multiple selectively advantageous haplotypes.[64] To scan selective sweeps in five Chinese *Pst* populations (G2, G3, G5, G6, and G7; see Results), we used two complementary methods: the composite likelihood ratio (CLR) method for hard sweeps and the integrated haplotype score (iHS) method for soft sweeps. The CLR is a composite likelihood approach to compare the likelihood of a model of positive selection to a model of neutral evolution, given that the allele frequency spectrum of the population under positive selection will be different from that expected under neutral evolution.[65] Prior to scanning, haplotype phasing was performed using SHAPEIT4.[66] First, the CLR test based on the site frequency spectrum (SFS) was implemented using SweeD, as this method allows the detection of intrapopulation hard sweeps.[65,67] This approach increases the robustness of demographic models because the null model relies on the variation of the sequence SFS across the whole genome rather than the standard neutral model.[65,68] To determine the threshold for selection analysis, we computed the distributions of CLRs in 1000 datasets per population, simulated under the inferred demographic models (see section 'Demographic reconstruction'). To reduce computation burden, only one chromosome with 6 Mb was simulated. The time and population size were scaled to $4N_0$ and N_0, respectively. The growth rate α at time t was calculated according to $N(t) = N_0 * \exp^{-\alpha t}$, where N_0 represents the initial population size. Five to eight timepoints from the best demographic history model were used in the simulation by setting -eN and -eN parameters in MacS.[69] The CLRs were generated for each simulated dataset using SweeD. In addition, we searched soft sweeps using the iHS test with the R package REHH2.0.[70] The iHS test compares extended haplotype homozygosity of ancestral alleles to that of derived alleles for each SNP to detect lengthy haplotype blocks. We set the maxgap option to 20,000 and limehh to 0.05 while using scan_hh. Outliers within 5 kb were regarded as the same selective sweep area. We obtained the selective sweep region by adding 5 kb at the flanks of each outlier site. To mitigate the risk of false positives in the CLR test due to low SNP density, we computed SNP density in 50-kb nonoverlapping windows along the whole genome. Only CLR outlier sites located in windows with an SNP density exceeding 100 were retained. Gene Ontology (GO) terms were assigned to the genes within the sweeps by searching the Pfam, SUPERFAMILY, TIGRFAM, SFLD, CDD, PIRSR, and PANTHER databases. Enrichment analyses were performed using the clusterProfiler (v4.6.0) package in R.[71]

2.6 | Demographic reconstruction

We exploited SMC++ software to construct the demographic history of Chinese *Pst* populations based on whole-genome sequences.[72] SMC++ combines the SFS and the coalescent hidden Markov models to calculate changes in effective population size over time, which makes the inference computationally efficient and accurate. SMC++ is robust to phasing errors, and it is recommended to use sample sizes of 2–10 individuals with high sequencing coverage in practice.[72] To distinguish missing regions from very long runs of homozygosity, the raw variants were filtered by setting –minGQ 60 using VCFtools, and low-quality variants were masked as missing, following SMC++ recommendations (https://github.com/popgenmethods/smcpp).[56] The SMC format was generated from the first dataset using smcpp vcf2smc, which was then used to estimate the effective population size with smcpp estimate. For each population, four isolates were randomly selected as distinct lineages for vcf2smc (specified by the -d option). We set the mutation rate at $2e^{-8}$ per base pair per generation according to the study of Xia et al.[73] Finally, the smcpp plot was used to plot the effective population size over time. To convert coalescent units from generations to years, we considered the epidemiology of Gansu Province, where *Pst* could oversummer and overwinter, serving as the major source of inoculum for Eastern China. In the Gansu Province, there is generally one generation before *Pst* overwinters. In the following spring, sporulation starts in March and lasts until June, with one generation per month. Therefore, five generations per year were used in the demographic history inference.

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2.7 | Gene function annotation and enrichment analysis

Genes detected in the selective sweep regions were functionally predicted using InterProScan v5.51–85.0.[74] We used a local lookup to categorize these genes into protein families (Pfam), Gene Ontology (GO) terms, SUPERFAMILY, TIGRFAM, SFLD, CDD, PIRSR, and PANTHER. We also predicted signal peptide and transmembrane signatures with SignalP 4.1 and TMHMM 2.0, respectively.[75,76] Proteins containing a signal peptide predicted using SignalP were defined as secreted proteins (SPs), after excluding those predicted as transmembrane using TMHMM.

To detect the phenotypes of mutants, we aligned the SPs to the Pathogen-Host Interactions database (PHI-base) version 4.10, released on 2 November 2020, using BLASTP 2.9.0.[77] We performed GO enrichment analyses for all genes located within the selective sweep regions detected by CLR and iHS. Enrichment analyses of GO terms were performed using the R package clusterProfiler, with a cutoff (p-value) set at < 0.05.[78] Only GO terms with at least five different genes assigned in the reference genome were included. We also searched effector protein sequences of known function in the CYR34 reference genome to identify effectors residing in the selective sweep regions.

3 | RESULTS

For the 69 Chinese *Pst* isolates, between 38.16% and 78.73% of the reads were mapped to the reference genome of CYR34, with sequencing depths ranging from 24.38× to 47.07× (Table S1). The unmapped reads were potentially thrips, plant hosts, or bacteria. Similar variations in the genomic read mapping rates of rust fungi (e.g., *Pgt*, the wheat stem rust fungus) have been recorded.[79,80] Additionally, sequencing data of 28 isolates from Australia, India, Ethiopia, Denmark, France, the UK, and the USA, all with mapping coverage greater than 20×, were retrieved and used in subsequent analyses (Table S2). After quality control and filtering, a total of 783,514 biallelic SNPs were called across all 97 isolates for further analyses, except in the population structure analysis in which 46,152 SNPs were used after thinning by 1 kb.

3.1 | Population structure and diversity

We inferred the population structure with phylogenetic analysis and ADMIXTURE based on whole-genome diversity data (Figure 1). The optimal number of clusters identified by ADMIXTURE was $K = 7$ (Figure S1). We defined a strain's cluster based on the criterion that its ancestry proportion was greater than or equal to 70%. If no clusters met this criterion, the strain was defined as an admixture. Overall, the phylogenetic clades and ADMIXTURE clusters revealed similar population structures. As shown in Figure 1, phylogenetic clade I and ADMIXTURE clusters G1 and G4 exclusively contain non-Chinese isolates, clade II shares the same composition as G6, isolates in clade IV are mostly composed with G2, clade V isolates primarily exhibit G7 ancestry, and clusters G3 and G5 collectively form clade III. Notably, almost all (26 out of a total of 28) non-Chinese isolates were exclusively grouped into clade I with no isolates from China. Additionally, the four non-Chinese admixture isolates exhibited a different ancestry composition pattern, containing G4 ancestry, compared to the admixture isolates from China. This distinction was more apparent when $K = 8$ (Figure S1). Given that only two out of the total of 28 non-Chinese isolates had a similar ancestry composition to the isolates from China, we speculated that there might be limited gene flow between *Pst* isolates from China and non-China. The PCA analysis showed a similar clustering pattern, in which *Pst* isolates from China were grouped into two clusters and separated from the remaining non-Chinese isolates (Figure 2A). Furthermore, two European isolates, DK0911 and 11/08, were clustered into G3. These two isolates have been identified as the Warrior race in recent research.[42,46] The "Warrior" race emerged in 2011 in Europe with greater virulence than the earlier isolates such as 78/66 and 08/21. Significantly, it has also been proved that the "Warrior" race shows close relatedness with isolates sampled in the near-Himalayan region.[18] Therefore, even though limited gene flow between Chinese and non-Chinese isolates, there is a possibility that migration events from Himalayan regions to European isolates occurred before 2011, contributing to increased diversity in Europe.[42]

For *Pst* isolates in China, ADMIXTURE analysis showed regional differentiation when $K \geq 4$ (Figure S1). The *Pst* populations from HB, SC, SX, and GS exhibited similar ancestry, whereas those from AH, YN, GZ, and CQ shared similar ancestral components (Figure 1). In general, a greater number (> 4) of assigned clusters indicated a more complex ancestral composition. In the ADMIXTURE analysis, the Chinese *Pst* isolates displayed a higher number of clusters, whereas the ancestry of non-Chinese isolates remained relatively stable.

We next attempted to illustrate the correlation between epidemiological regions and gene clustering in *Pst* populations (Figure 2B). The clusters showed the relevance of the epidemic regions in China. We generated a Sankey plot to examine the genetic composition of the populations from each epidemiological region and province. As shown in Figure 2B, G2 is detected in the majority of Chinese provinces, mainly in the AutS and SprE regions. G3, G5, and G7 are

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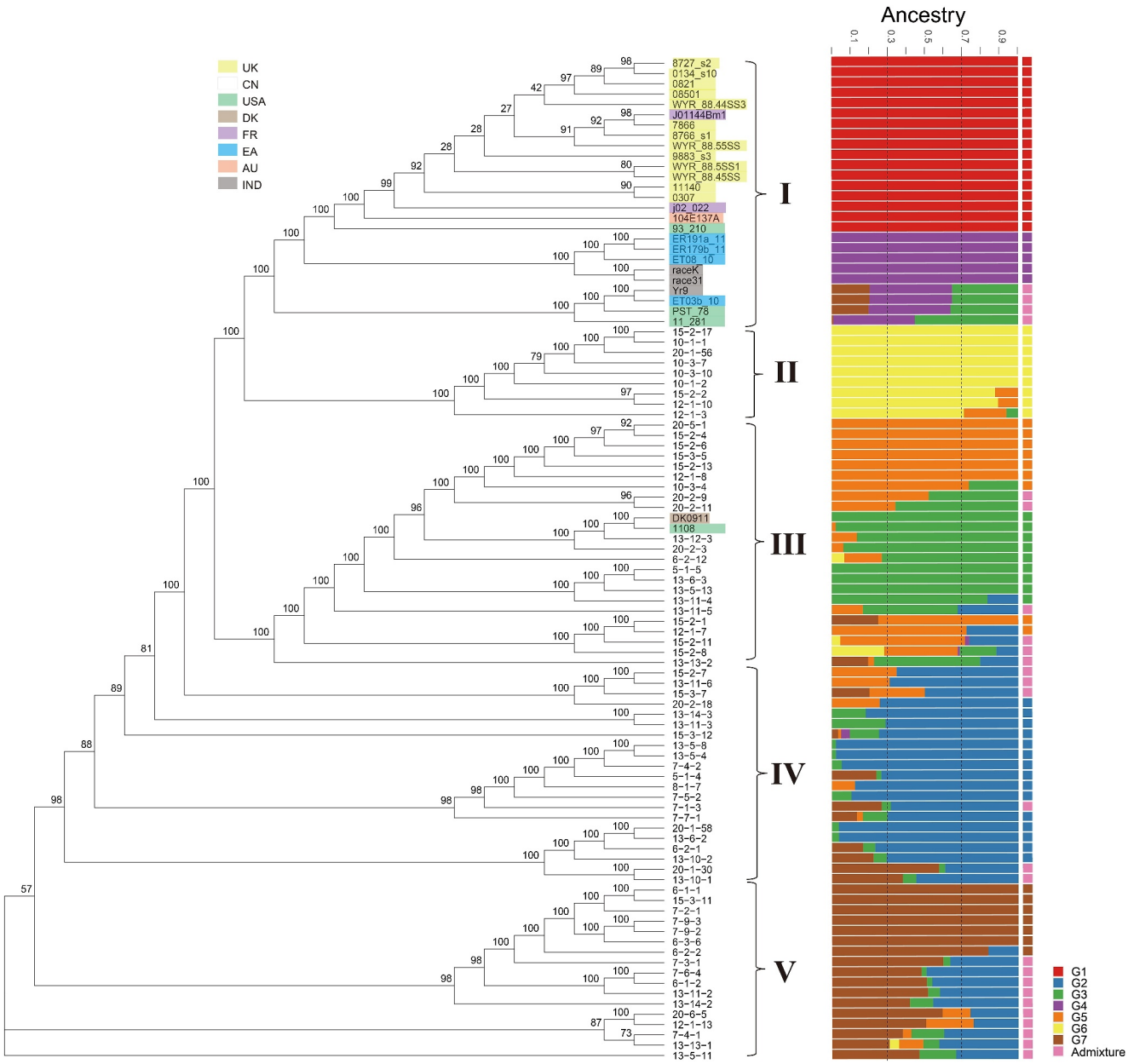

FIGURE 1 Sampling information, phylogenetic relationships, and population structure of the *Puccinia striiformis* f. sp. *tritici* (*Pst*) isolates used in this study. Phylogenetic relationship of a total of 98 *Pst* isolates is used in this study. The neighbor-joining tree was constructed based on genetic distances estimated using genome-wide SNPs. For visualization purposes, the tree branches were scaled to the same length but were not proportional to the mutation rate. The numbers around the phylogenetic tree nodes represent bootstrap values. The population structure is inferred from ADMIXTURE at the optimal number of 7 clusters. The isolates were assigned to an admixture cluster when their highest ancestry to a specific cluster was below 0.7. AU, Australia; CN, China; DK, Denmark; EA, Eastern Africa; FR, France; IND, India; UK, United Kingdom; USA, United States of America.

primarily distributed in the SprE, SprS, and AutS regions, respectively. The G6 population is detected in SprE and SprS, but not in AutS. In contrast, *Pst* populations from the AutS region are grouped into G2, G3, and G7, whereas those from SprS are mainly grouped into G5 and G6, indicating divergence and differences in the genetic composition of these two epidemiological regions. Notably, the *Pst* population from Chongqing (belonging to SprS) was grouped into four genetic clusters. Interestingly, although both Anhui and Hubei belong to the SprE region, the isolates from Anhui had similar ancestry to isolates from the SprS region (e.g., Yunnan and Guizhou provinces), whereas the isolates from Hubei share more ancestry with isolates from the AutS region (e.g., Gansu Province), suggesting different initial inocula in different SprE regions.

We calculated nucleotide diversity (θ_π) of each ADMIXTURE-defined population (Figure 3A). The

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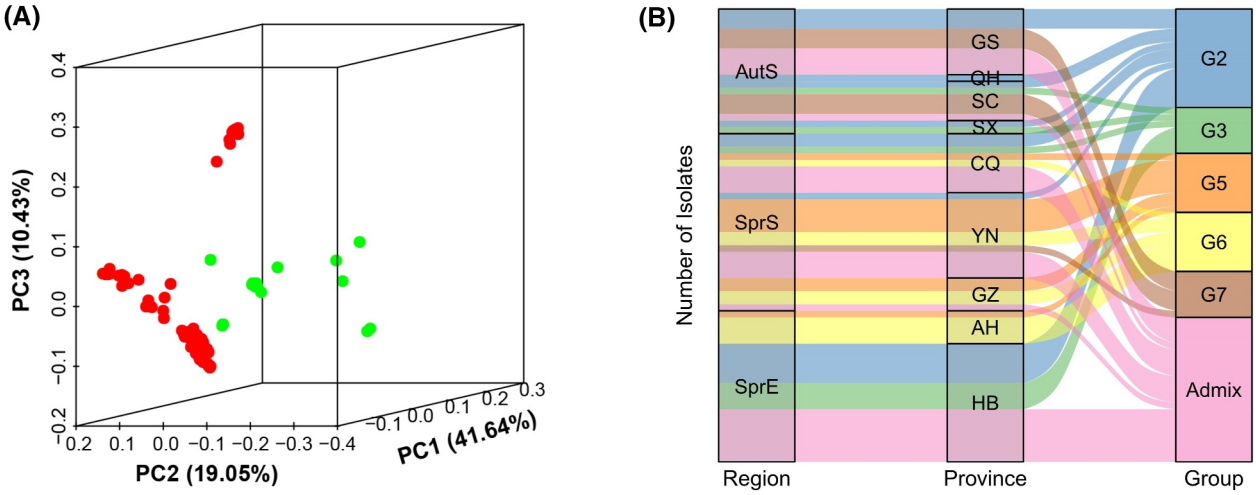

FIGURE 2 Clustering of *Puccinia striiformis* f. sp. *tritici* isolates. (A) The 3D scatter plot of the principal component analysis of 98 worldwide *Pst* isolates using whole-genome-wide SNPs. The isolates from China and non-China are represented by solid red and green circles, respectively. The first three principal components account for 41.64%, 19.05%, and 10.43% of the overall variance. (B) The Sankey plot of 69 *Pst* isolates from China across epidemiological regions, provinces, and genetic clusters determined by ADMIXTURE. AH, Anhui; AutS, autumn sources of inoculum; CQ, Chongqing; GS, Gansu; GZ, Guizhou; HB, Hubei; QH, Qinghai; SC, Sichuan; SprE, spring epidemic areas; SprS, spring sources of inoculum; SX, Shanxi; YN, Yunnan.

average θ_π ranged from 0.0035 to 0.0063, with the two *Pst* populations from China (G3 and G5) and one international population (G4) exhibiting the lowest θ_π compared to the remaining populations. In fact, these three populations had the lowest number of biallelic SNPs (Figure 3A). This low level of nucleotide diversity in the G3, G4, and G5 populations was observed throughout the entire genome (Figure 3A). The alignment results did not display biased sequencing coverage or mapping quality issues for the isolates in these populations (Tables S1 and S2). We speculated that the low nucleotide diversity of these three populations might be attributed to the relatively small long-term effective population size (refer to "Demographic history of *Pst* populations") and strong bottleneck effects. For instance, G3 populations (primarily from Hubei Province) could not oversummer and rely heavily on a large amount of inocula for autumn and spring epidemics every year.

The average Tajima's *D* values calculated using 10-kb genomic windows along the genome were all positive but varied among different *Pst* populations. The international *Pst* group G4 had the lowest average Tajima's *D* value of 0.5940, whereas the other international group G1 had the highest value of 3.8823 (Figure 3B). Similar to the case of nucleotide diversity, the low Tajima's *D* value in G4 was also observed throughout the entire genome (Figure 3B). The variation in Tajima's *D* values suggests that different selection pressures have been exerted on the *Pst* genome across the various populations.

Given the presence of alternate host species and the potential for sexual reproduction in China, we compared the LD between the Chinese *Pst* population and the non-Chinese population. As expected, the LD coefficients (R^2) decreased with increasing physical distances between sites, exhibiting similar decay patterns for both Chinese and international *Pst* populations (Figure 4). For comparison, we also calculated R^2 for our previously established *Pst* selfing population.[60] Our findings revealed a significantly slower LD decay in the selfing population, resulting in R^2 values that remained much higher than those of both Chinese and non-Chinese *Pst* populations. The slower LD decay was expected, as only one cycle of sexual reproduction occurred in the selfing population. Furthermore, we retrieved a natural population of *Pgt*, the causal fungus of wheat stem rust disease, with 77 isolates, and estimated the R^2. The results showed that *Pgt* had a much quicker rate of LD decay (R^2 reduced to 0.25 within <1 kb) compared to *Pst* populations (R^2 reduced to 0.5 within > 5 kb) (Figure 4). Taken together, these findings suggest that *Pst* populations have undergone sexual reproduction in nature, albeit at a much lower frequency than that observed in their sister species, *Pgt*. At this point, the investigation of somatic and/or parasexual reproduction might be more interesting for understanding the evolution of *Pst*.

3.2 | Genome scan of selective sweeps in Chinese *Pst* populations

To identify selective sweeps in Chinese *Pst* populations, we compared the CLR distributions from real populations with those generated from simulated

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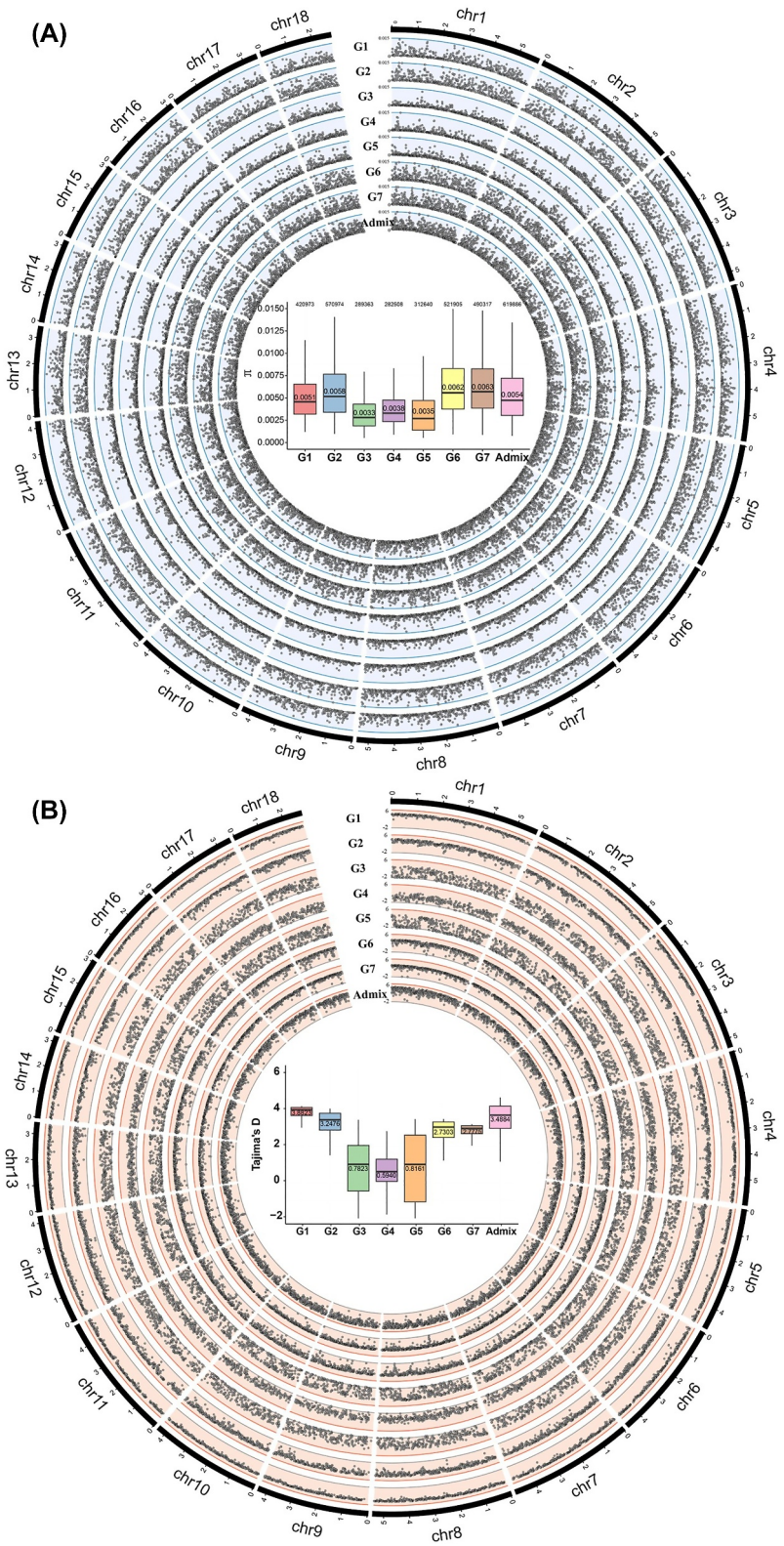

FIGURE 3 Population diversity of *Puccinia striiformis* f. sp. *tritici* isolates. (A) Nucleotide diversity (θ_π) of ADMIXTURE-defined *Pst* populations. The circos plot illustrates the distribution of θ_π along the chromosomes for each population, whereas the boxplot shows the summary of θ_π for each population. Solid lines in the boxplot represent the mean, and numbers above each boxplot represent the total number of biallelic SNPs detected in the population. (B) Tajima's *D* of ADMIXTURE-defined *Pst* populations. The solid line in the boxplot represents the mean, and the numbers above the boxplots indicate the total number of biallelic SNPs detected in the population.

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datasets. The 99.5th percentiles of CLR from real populations ranged from 50.33 to 286.15, whereas those from each simulated dataset ranged from 3 to 22 (Figure S2). A similar pattern was also reported in *Pgt* populations.[61] Therefore, the empirically defined threshold of 99.5% was more conservative and should reduce the rate of false positives in the selection analyses. By applying this threshold, 23–97 regions harboring hard sweeps were identified in five ADMIXTURE-defined Chinese *Pst* populations (Table 1). These regions, which accounted for 3.98% of the whole genome, were found across all 18 chromosomes (Figure 5). The lengths of individual regions ranged from 5.0 to 76.8 kb.

We also performed iHS statistics to detect soft sweeps using a threshold of 99.5%. In populations G2, G3, G5, G6, and G7, we detected a total of 209, 158, 57, 61, and 65 selective sweep regions, respectively, that have experienced recent positive selection. Each of these regions varies in length from 10.0 to 21.7 kb, collectively spanning 3.95% of the genome (Figure 6).

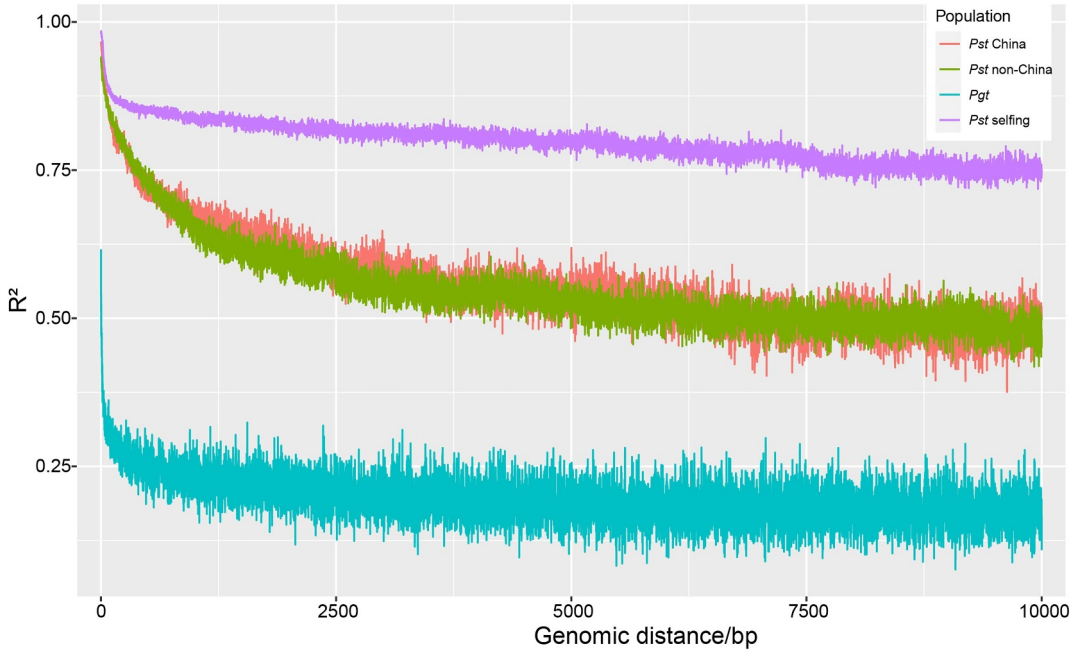

FIGURE 4 Linkage decay of the *Pst* populations and comparison to *Puccinia graminis* f. sp. *tritici* (*Pgt*), the causal agent of stem rust in wheat. Red lines, the *Pst* population in China; green line, the international *Pst* population excluding Chinese isolates; purple line, the selfing population; blue line, the *Pgt* population with 77 isolates reported by Guo et al.[61]

TABLE 1 Summary statistics of the selective sweep regions identified in Chinese *Pst* populations.

Detection method	Population	Number of selective sweep regions	Total length of regions affected by sweep		Number of genes in all sweep regions
			Accumulative size (kb)	Percentage of whole genome (%)	
CLR	G2	44	581.9	0.74	112
	G3	23	504.4	0.64	100
	G5	56	676.7	0.86	195
	G6	35	572.9	0.72	131
	G7	97	808.8	1.02	183
iHS	G2	209	1196.3	1.51	376
	G3	158	870.9	1.10	271
	G5	57	320.3	0.40	106
	G6	61	367.1	0.46	120
	G7	65	371.4	0.47	115

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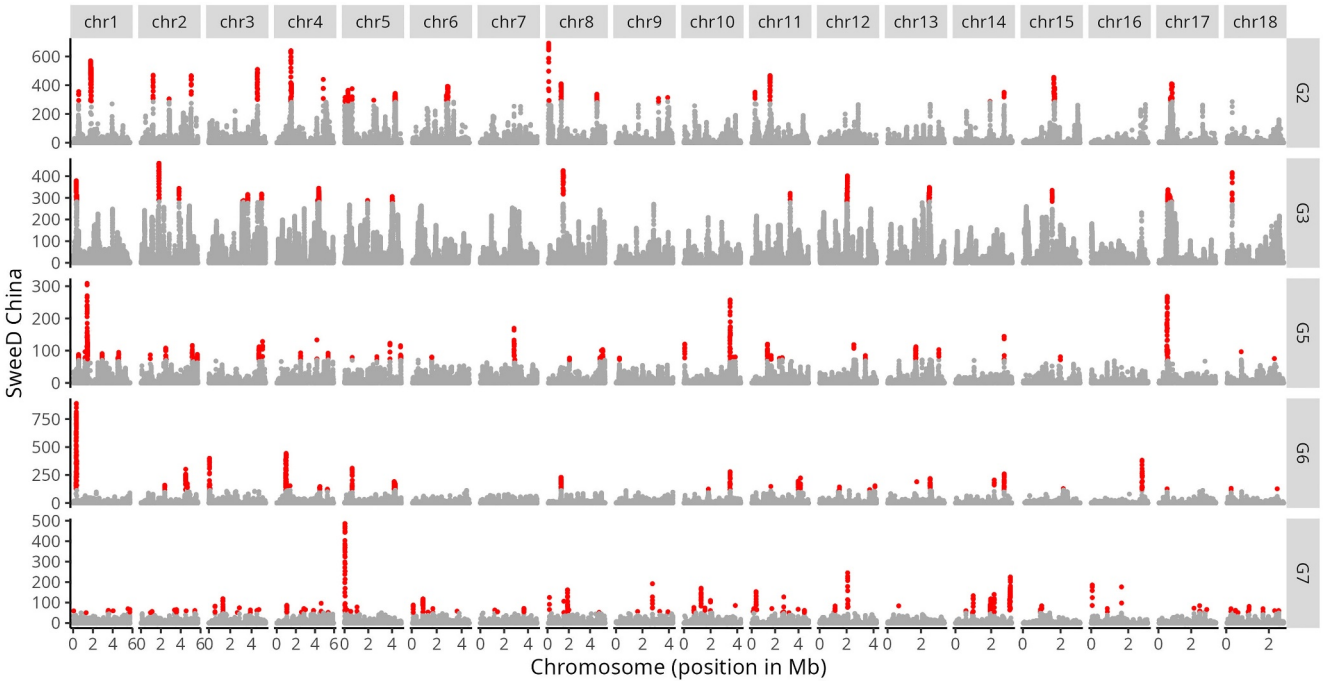

FIGURE 5 Genome-wide hard sweeps in five Chinese *Pst* populations detected by composite likelihood ratio (CLR). Each column represents a specific chromosome, whereas each panel represents a distinct group. The red dots indicate outlier sites over the 99.5% threshold.

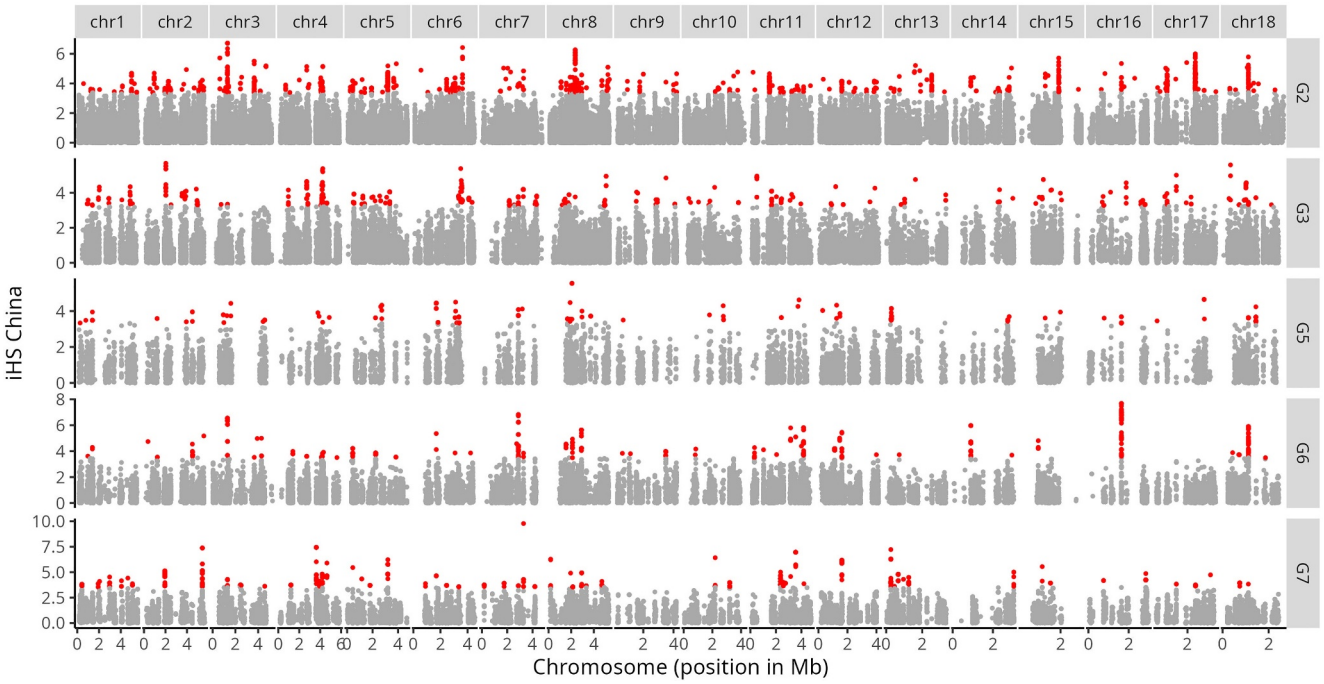

FIGURE 6 Genome-wide soft sweeps in five Chinese *Pst* populations detected by integrated haplotype score (iHS). Each column represents a specific chromosome, and each panel corresponds to a group, with red dots denoting outlier sites exceeding the 99.5% threshold.

3.3 | Gene function in selective sweep regions

After annotation, we detected 1526 and 988 genes in selective sweep regions identified by CLR and iHS, respectively. Genes encoding SPs are estimated to constitute 11.4% and 10.5% of the total, which is higher than the genome-wide average of 9.7% (Table 2). Furthermore, 33.3% and 33.6% of the SP genes in selective sweep regions are predicted to be effector

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genes, which also exceeds the genome-wide average of 32.6%. A large number of genes within these sweeps have potential homologs and are predicted to exhibit reduced virulence when mutated, suggesting that they are involved in pathogen-host interactions (Table S3).

GO enrichment analysis showed that the genes within soft sweeps were enriched in chitin synthase activity (GO:0004100), hexosyltransferase activity (GO:0016758), ribosome (GO:0005840), and other functions (Table 3). We also found that the soft sweeps were enriched in various binding activities, including 48 genes associated with protein binding activities (GO:0005515), 32 genes related to nucleic acid binding (GO:0003676), 25 genes linked to zinc ion binding (GO:0008270), and 21 genes involved in DNA binding (GO:0003677) (Table S4 and Figure S3). Furthermore, genes within the iHS selective sweep exhibited mRNA-related functions, such as translation initiation (GO:0006413), mRNA splicing via spliceosome (GO:0000398), and regulation of transcription by RNA polymerase II (GO:0006357). Similar to the gene functions observed in the soft sweeps, the hard sweeps also contained a large number of genes associated with binding activities (Tables 3 and S5, and Figure S4).

In addition, several known functional effectors were identified to be located in the selective sweep regions, such as *PNPi* (AOS49611.1), *PsKPP4* (AXL48515.1), Ras-like C3 botulinum toxin substrate 1 (KNE95594.1), and PGT-hesp-767 (XP_003322973.1) in the soft sweeps, as well as Ras-like protein Rab7 (KNF04400.1) in the hard sweeps.

TABLE 2 Content of genes, predicted secreted proteins, and predicted effectors in the whole genome and selective sweep regions detected by CLR and iHS.

Region	Detection method	Number of genes	Gene density (per 10 kb)[a]	Number of secreted proteins	Secreted protein ratio (%)[b]	Number of effectors	Effector ratio (%)[c]
Selective sweep regions	CLR	1526	2.36	174	11.40	58	33.3
	iHS	988	3.16	104	10.52	35	33.6
Genome		16,455	2.1	1602	9.7	523	32.6

[a]Gene density was calculated by dividing the number of genes by the length of total regions.
[b]Secreted protein ratio was calculated by dividing the number of secreted proteins by the number of genes.
[c]Effector ratio was calculated by dividing the number of effectors by the number of secreted proteins.

TABLE 3 Gene Ontology functional enrichment of genes in selective sweep regions detected by CLR and iHS (p-value < 0.05).

Methods	GO ID	GO description	*p*-value	Ontology[a]
CLR	GO:0004100	Chitin synthase activity	1.78E-02	MF
	GO:0006412	Translation	3.01E-02	BP
	GO:0003735	Structural constituent of ribosome	4.46E-02	BP
	GO:0004222	Metalloendopeptidase activity	4.93E-02	BP
	GO:0016758	Hexosyltransferase activity	1.38E-02	CC
	GO:0005840	Ribosome	4.32E-03	CC
	GO:0000160	Phosphorelay signal transduction system	2.07E-02	MF
	GO:0046982	Protein heterodimerization activity	2.98E-02	MF
iHS	GO:0000398	mRNA splicing via spliceosome	2.25E-03	BP
	GO:0016226	Iron-sulfur cluster assembly	2.30E-02	BP
	GO:0005839	Proteasome core complex	4.22E-02	BP
	GO:0006357	Regulation of transcription by RNA polymerase II	4.42E-02	BP
	GO:0046873	Metal ion transmembrane transporter activity	1.28E-03	MF
	GO:0030001	Metal ion transport	1.93E-03	MF
	GO:0006413	Translational initiation	8.62E-03	MF
	GO:0051603	Proteolysis involved in protein catabolic process	4.22E-02	MF
	GO:0004185	Serine-type carboxypeptidase activity	5.00E-02	MF

Abbreviations: BP, biological processes; CC, cellular components; MF, molecular function.
[a]Gene Ontology terms.

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3.4 | Demographic history of *Pst* populations

We attempted to infer the historical effective population sizes of different *Pst* populations. All isolates listed in Tables S1 and S2 are combined to represent the international *Pst* population (G_*Pst*), and isolates in Table S1 are used to represent the Chinese *Pst* population (G_CN). In Figure 7A, we present the inferred population size history spanning from 20,000 to 200 years ago. During the period from 20,000 to 13,000 years ago, the population sizes of G_CN and G_*Pst* experienced a highly similar dynamic and remained stable. Both populations experienced a steep decline in size from 13,000 to 10,000 years ago (Figure 7A), followed by a short period of stability. Subsequently, the two populations began to diverge, followed by a round of size expansion and reduction from 3000 to 60 years ago. The two lowest points of effective population sizes suggest that the pathogen populations are experiencing bottleneck events.

To reveal more details of *Pst* demography, we further inferred the size histories for each ADMIXTURE-defined population. As shown in Figure 7B, all populations experienced similar dynamics to those of G_*Pst* and G_CN in the distant past, characterized by a steep decline starting 30,000 years ago. The populations diverged around 10,000 years ago, with a slower decline in population sizes until about 3000 years ago. From 2000 to 3000 years ago, the G1, G2, G6, and G7 populations began to deviate from the remaining three subpopulations, with a slight increase in population size until 800 years ago, followed by a steep decline up to the very recent past. Interestingly, the inferred demographic histories of G3 (mainly from Hubei, China), G4 (from India and Ethiopia), and G5 (from Yunnan, China) displayed similar dynamics, with a moderate decline in population size following their divergence from other subpopulations approximately 300–400 years ago. Furthermore, we noted that most *Pst* populations have been experiencing a decline in size over the past 100 years, which might reflect the directional selection of the pathogen due to modern breeding programs.

4 | DISCUSSION

4.1 | Limited gene flow between *Pst* populations from China and other countries and low frequency of sexual reproduction in natural *Pst* populations

Our population genomic analyses unraveled several key features of the global *Pst* population. First, Chinese *Pst* populations exhibited limited admixture with those from other countries, suggesting that the pathogen might have evolved independently in China, given that the large geographical area and diverse cropping systems provide both overwintering and oversummering places for *Pst*.[8,81] Second, international *Pst* populations displayed similarly high levels of diversity compared to Chinese populations. Previous studies based on tens of microsatellite markers suggested that populations from Europe, Australia, Eastern Africa, India, and the USA displayed low diversity, whereas populations from China were highly diverse.[21] We speculate that this inconsistency might stem from the technical bias of genotyping techniques, as most microsatellite markers were initially developed and optimized for specific populations, whereas genome-wide SNPs could minimize such biases. Third, our analyses indicate that both Chinese and international *Pst* populations have the potential for sexual reproduction. Even though the sexual life cycle of *Pst* was discovered a decade ago, the extent of sexual reproduction under natural

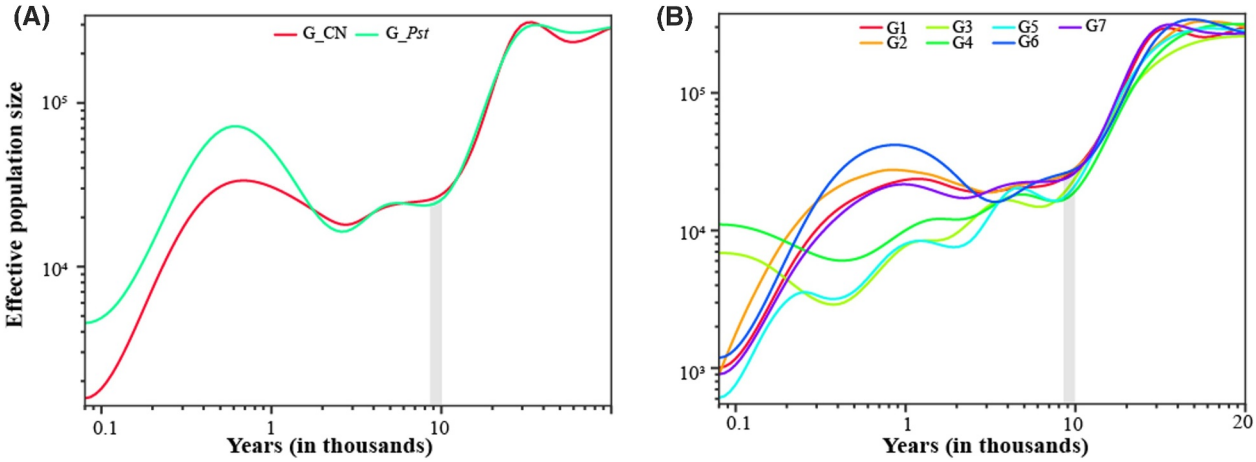

FIGURE 7 Effective population size of *Pst* populations over time. Each line represents a population. (A) Worldwide *Pst* population (G_*Pst*) compared to the combined Chinese *Pst* population (G_CN). (B) All Chinese *Pst* populations.

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conditions remains unclear.[9] Previous analyses based on several molecular markers could only infer the sexuality of the loci from which the markers were developed. For example, Ali et al., using 20 microsatellites, demonstrated that the population of *Pst* on a continental scale was clonal except in the Himalayan and near-Himalayan (China) regions.[21] To our knowledge, this is the first study to use genome-wide loci to assess LD in *Pst*. Moreover, we took advantage of an established selfing population with theoretically one meiosis as a control for comparison with the natural population. On one hand, the LD decay in both Chinese and international *Pst* populations was much faster than that in the selfing population, suggesting that natural *Pst* has accumulated extensive recombination. On the other hand, the LD decay of *Pst* detected in the present study was slower than that in its sister species, *P. graminis* f. sp. *tritici*, which undergoes frequent sexual reproduction (Figure 4), and other fungi (e.g., *Serpula lacrymans*).[39] This finding confirms our previous speculation that the frequency of sexual reproduction in *Pst* is low in nature. The low frequency of sexual reproduction implies that the importance of alternate hosts in *Pst* epidemics should not be overestimated.

Historically, three main *Pst* epidemiological regions were designated in China, namely, the SprS, AutS, and SprE.[8] We did not establish clear correspondences between these epidemic areas or provinces and genetic populations. This might be attributed to the high mixture of ancestry in each isolate, which was possibly caused by migration among epidemiological regions/provinces. The low germination rate of teliospores and the rarity of sexual reproduction in nature in China further support this explanation, as indicated by our LD analysis (Figure 4).[82,83] In the AutS, Gansu and its neighboring areas were supposed to be the oversummering region and the origin area of Chinese *Pst* with a high diversity and recombination rate.[29] Traditional epidemiology considered that *Pst* isolates in Gansu exhibit high polymorphism, whereas we found that isolates from autumn sources were mainly grouped into G2 and G7. In contrast, the spring sources mainly contained G5 and G6 populations. The SprE comprised all five populations, reflecting the impact of both SprS and AutS. Recently, Li et al. proposed that Yunnan was another origin of Chinese *Pst*, due to its geographical proximity to the Himalayans.[84] In contrast to the findings of Li et al., we detected the lowest nucleotide diversity in G5, which is predominantly distributed in Yunnan.[84]

4.2 | Extensive hard and soft sweeps associated with *Pst* adaptation

Our results indicated that *Pst* adaptation is associated with both hard and soft selective sweeps. The genome-wide selective signals in *Pst* accounted for 3.95% and 3.98% of the genome. This pervasive selection has also been detected in other agricultural plant pathogens,[28,30,31] such as 0.5%–4% selective sweep regions in another wheat fungal pathogen, *Zymoseptoria tritici*. The classical model of positive selection suggests that the time to adaptation is proportional to the product of the mutation rate of the adaptive variant and the effective population size.[7] Based on this, given the mutation rate of 10^{-8} per site per generation and the effective population size of 10^5, the time to adaptation for *Pst* would be of the order of thousands of generations, which is much slower than the observed (i.e., emergence of new races within only years to decades). One explanation for this conflict is that the assumption that adaptation involves a single adaptive variant (i.e., hard selective sweep) is not valid in the classical model. Gladieux et al. proposed that adaptation may be faster given larger mutational targets (i.e., a large number of base pairs within an allele, corresponding to soft selective sweeps).[7] Indeed, our results of balanced hard and soft sweeps in *Pst* are consistent with soft sweeps-associated rapid adaptation. Another explanation may be the underestimation of the effective population size. Most samples were collected from the most commonly grown commercial wheat, yet rust exists in small amounts on wild relatives or less commonly grown varieties. These populations, though not easily detected in field surveys, can quickly become epidemic under suitable conditions, implying a larger effective population size than is commonly assumed.

Effectors play a crucial role in host-pathogen interactions, which makes them undergo selection pressures exerted by hosts. For example, the filamentous pathogen *Melampsora lini* showed genetic variation at effector loci *AvrP123* and *AvrP4* due to host adaptation.[85] In a study of the wheat pathogen *Parastagonospora nodorum*, effectors also displayed differentiation driven by local adaptation.[86] Hence, we performed effector prediction in the *Pst* genome. A higher ratio of gene density, SP ratio, and effector ratio was displayed in selective sweep regions than in the whole genome across all populations, as determined by each testing method. The results imply that pathogenicity genes might undergo recent positive selection. For example, we performed sequence alignment of functionally known *Pst* effectors and detected *PNPi* and *PsKPP4* effectors in selective sweep regions. The PNPi effector interacts with wheat *NPR1*, leading to reduced induction in wheat during infection.[87] *PsKPP4* strengthens the pathogenicity of *Pst* by regulating infection-related morphogenesis.[88] For genes within the sweeps but with unknown functions, it would be valuable to explore the specific adaptive phenotypes associated with these genes.

In the selective sweep genes, we identified genes encoding major facilitator superfamily (MFS)

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transporters, sugar transporters, ATPase family associated with various cellular activities, and heat shock protein (*Hsp*70) family. MFS transporters are involved in pathogen sensitivity to fungal toxins and fungicides, pathogenicity, and response to oxidative stress.[89,90] Hartmann et al. detected abundant transmembrane transporter genes in selective sweep regions of continent-scale *Zymoseptoria tritici*, implying selection forces from the environment and host.[30] DNA repair proteins swi10 and rad10 in yeast exhibit homology to the human ERCC1,[91] which is related to the repairing of DNA damage caused by UV light.[92] Heat shock proteins play a part under stress conditions in fungi, with Hsp70 protein involved in the response to cold stress.[93] Additionally, temperature and light responses reflected selection forces from abiotic factors on *Pst* populations.

4.3 | Coevolution between *Pst* fungi and wheat domestication

Our demographic analysis revealed that changes in the effective population size of *Pst* populations coincide with key events in wheat domestication. The sizes of all *Pst* populations remained relatively stable before the beginning of hexaploid wheat formation, which is estimated to have occurred 9000–10,000 years ago.[94,95] Remarkably, strong bottleneck events were identified for all *Pst* populations during wheat domestication. These bottleneck events could be explained by the restricted genetic variability during wheat domestication.[96] Thus, this indicated that the demography of *Pst* was affected by the domestication of its host plant. Similar effects of crop domestication on pathogens have also been observed in other pathosystems, such as the maize smut pathosystem.[34] Although the trends of population sizes vary, we noticed a steady decline for all *Pst* populations in the past 200 years. Additionally, the breeding programs of wheat started during this period, with the priority of developing resistance to stripe rust.[96,97] This coincidence led us to speculate that the intentional breeding and widespread deployment of single/major resistance genes could continuously contribute to the decline in effective population sizes of *Pst*, even though such resistance could be easily overcome by the evolving virulence of *Pst*. However, further exploration is needed to determine whether the deployment of multiple resistance genes, including those from wild relatives, has impacts on *Pst* population demography, because a greater diversity of resistance genes may exert less selection pressure on pathogen populations.

In summary, through population genomic analyses of global *Pst* populations, we revealed unprecedented features within these populations. Our findings indicate a limited gene flow between *Pst* populations in China and other countries. Additionally, we identified widespread selective sweep regions across the *Pst* genome. Our results illustrate that the rapid adaptation of *Pst* may be associated with both hard and soft sweeps. Genes located within these selective sweeps are enriched for functions related to pathogen-host interactions and environmental adaptation. This study reinforces the view that fungal plant pathogens, characterized by primarily clonal reproduction and smaller effective population sizes, can exhibit widespread selective sweeps. Further research involving stripe rust pathogen populations from wheat wild relatives is necessary to illustrate the details of population evolution and to evaluate the impact of modern agriculture on changes in *Pst* populations. A comprehensive understanding of the coevolution between pathogens and host plants will enable the development of sustainable strategies to control the pace of pathogen evolution in agroecosystems.

AUTHOR CONTRIBUTIONS

Yun Xing: Formal analysis; investigation; writing—original draft; writing—review and editing. **Chongjing Xia:** Formal analysis; supervision; writing—review and editing; funding acquisition; conceptualization. **Liang Huang:** Methodology; validation; formal analysis; writing—review and editing. **Na Zhao:** Methodology; validation; investigation; writing—review and editing. **Hongfu Li:** Methodology; software; investigation; writing—review and editing. **Xingzong Zhang:** Methodology; software; investigation; writing—review and editing. **Age Qiu:** Methodology; investigation; validation; writing—review and editing. **Wanqiang Tang:** Methodology; validation; investigation; writing—review and editing. **Meinan Wang:** Methodology; investigation; validation; writing—review and editing. **Xianming Chen:** Methodology; validation; investigation; writing—review and editing. **Bo Liu:** Methodology; validation; investigation; writing—review and editing. **Hao Zhang:** Investigation; methodology; validation; writing—review and editing. **Li Gao:** Methodology; validation; funding acquisition; writing—review and editing. **Wanquan Chen:** Supervision; project administration; writing—review and editing. **Taiguo Liu:** Supervision; resources; project administration; conceptualization; writing—review and editing

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive in the National Genomics Data Center, China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA010026) that are publicly accessible at https://ngdc.cncb.ac.cn/gsa.

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